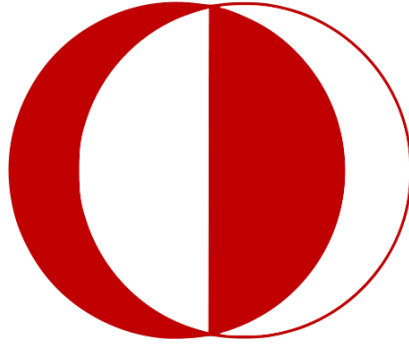




**Middle East Technical University**

## **EE568 Project-1**

Torque in a Variable Reluctance Machine

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# 1 Introduction

This report outlines the primary objectives and scope of the torque analysis for a variable reluctance machine. The study encompasses analytical derivations and computational verifications using COMSOL Multiphysics. Both analytical models and Finite Element Analysis (FEA) are employed to evaluate the magnetic circuit under linear and non-linear material conditions. Finally, a practical on-off control method is proposed to sustain continuous rotation based on the obtained torque profiles.

## 2 Analytical Modelling (Q1)

### 2.1 Reluctance, Inductance and Torque Calculations (Q1-a, Q1-b)

The maximum and minimum reluctance values of the variable reluctance machine were analytically calculated under the assumption that the core is infinitely permeable. This assumption simplifies the magnetic circuit to consist primarily of the air-gap reluctances. Based on the geometric dimensions provided, the maximum inductance at the aligned position ( $L_{max}$ ) was calculated as 0.0424 H, and the minimum inductance at the fully unaligned position ( $L_{min}$ ) was 0.00848 H. The inductance variation is assumed to follow a continuous sinusoidal profile as a function of the rotor angle  $\theta$ . The mathematical expression for the inductance is given by:

$$L(\theta) = L_0 + L_2 \cos(2\theta) \quad (1)$$

where the average inductance  $L_0$  and the inductance amplitude  $L_2$  are defined as:

$$L_0 = \frac{L_{max} + L_{min}}{2} \quad , \quad L_2 = \frac{L_{max} - L_{min}}{2} \quad (2)$$

Under a constant DC excitation current  $I$  of 2.5 A, the torque generated by the system is derived from the spatial derivative of the magnetic co-energy  $W_c$ . The co-energy in a linear magnetic system is expressed as:

$$W_c = \frac{1}{2} L(\theta) I^2 \quad (3)$$

The electromagnetic torque  $T_e$  is obtained by differentiating the co-energy with respect to the rotor position  $\theta$ :

$$T_e = \frac{\partial W_c}{\partial \theta} = -I^2 L_2 \sin(2\theta) \quad (4)$$

Using the calculated values, the analytical torque function demonstrates a theoretical peak value of 106 mN·m.

### 2.2 Method to Include Non-linear Effects (Q1-c)

To improve the analytical model, non-linear effects such as fringing flux and non-homogeneous flux distribution must be considered. Fringing at the air gaps increases the effective cross-sectional area, which effectively decreases the total air-gap reluctance and alters the inductance curve. Carter's coefficient can be utilized to mathematically account for these fringing fields in analytical equations without performing full finite element analysis.

### 3 FEA Modelling: 2D Linear Materials (Q2)

#### 3.1 Flux Density Vectors (Q2-a)

The variable reluctance machine was modeled in COMSOL Multiphysics using a linear soft iron material with a constant relative permeability ( $\mu_r$ ) of 2500. The flux density vectors for the rotor positioned at 0 degrees, 45 degrees, and 90 degrees were evaluated. Figure 1 and Figure 2 present the aligned and partially unaligned positions, respectively. Figure 3 demonstrates the completely unaligned position at 90 degrees. In this unaligned state, the reluctance of the magnetic circuit reaches its maximum due to the large effective air gap, resulting in a significantly lower and dispersed magnetic flux density across the core compared to the fully aligned position.

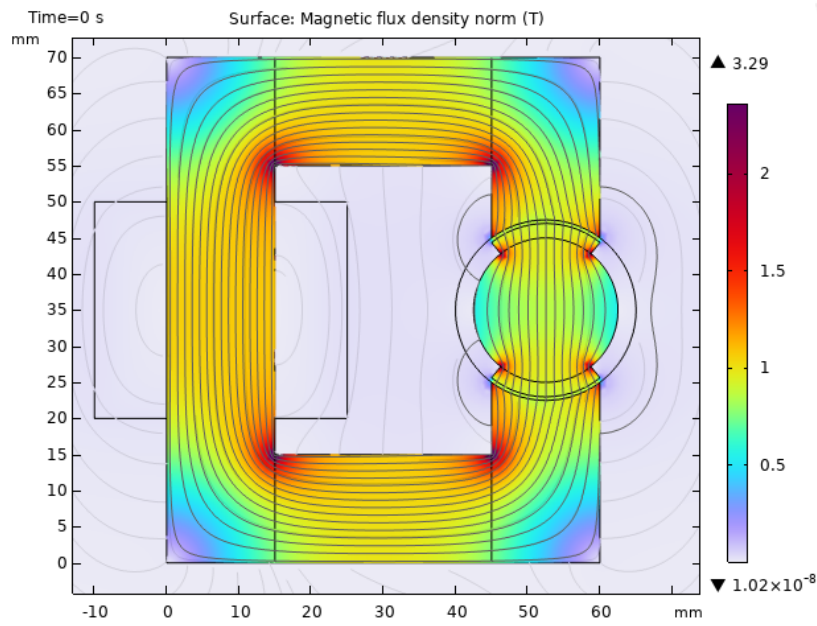


Figure 1: Flux density vectors at 0 degrees (aligned position) for linear material.

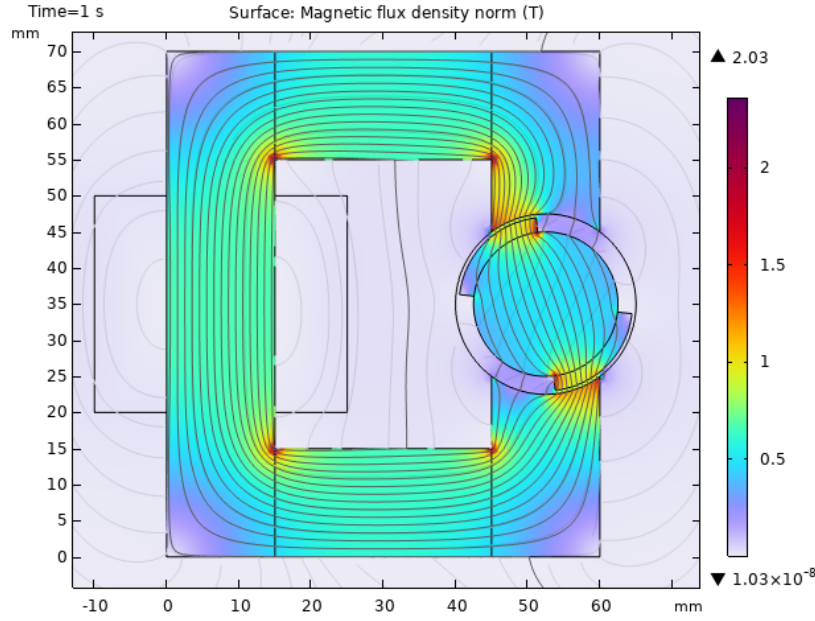


Figure 2: Flux density vectors at 45 degrees for linear material.

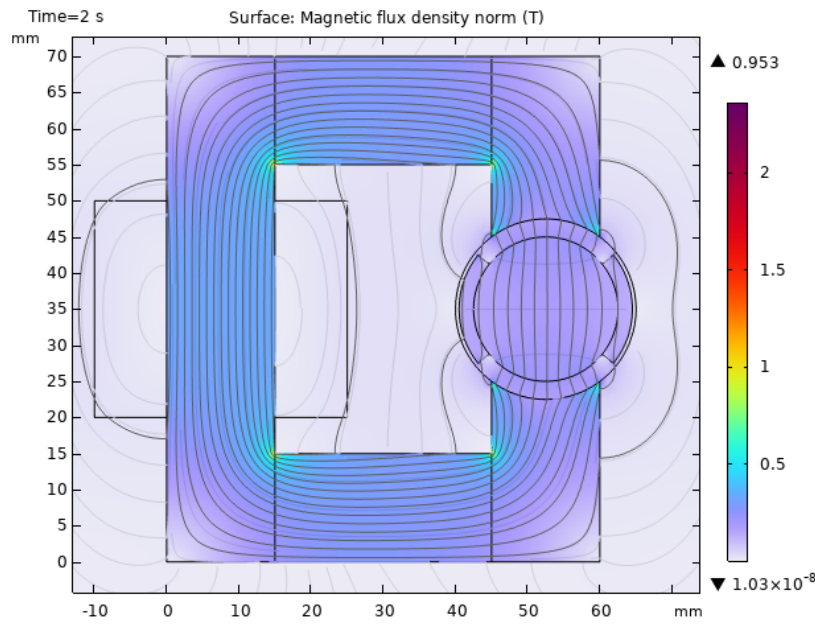


Figure 3: Flux density vectors at 90 degrees (unaligned position) for linear material.

### 3.2 Inductance, Stored Energy, and Torque (Q2-b, Q2-c)

Through the global evaluation of the 2D linear model, the inductance, stored magnetic energy, and axial torque were extracted. At the fully aligned position (0 degrees), the stored magnetic energy was computed as 0.149 J with an inductance of 0.0477 H. To satisfy the evaluation criteria and accurately capture the system dynamics, data was collected for at least ten operational positions. Table 1 details these extracted torque values.

Table 1: Axial torque values at discrete rotor positions for the linear material model.

| Rotor Angle (°) | Axial Torque (mN·m) |
|-----------------|---------------------|
| 0.0             | 0.15                |
| 22.5            | -78.44              |
| 45.0            | -86.08              |
| 67.5            | -68.97              |
| 90.0            | -0.24               |
| 112.5           | 69.64               |
| 135.0           | 86.64               |
| 157.5           | 79.08               |
| 180.0           | 0.16                |
| 360.0           | 0.15                |

To validate the models, the computational results obtained from the COMSOL simulations were plotted against the theoretical analytical derivations. The idealized analytical model assumes a perfect sinusoidal variation and predicts a higher peak torque. The FEM results accurately capture the flux leakage and fringing fields, resulting in a simulated peak torque of 86.64 mN·m at the 135-degree position.

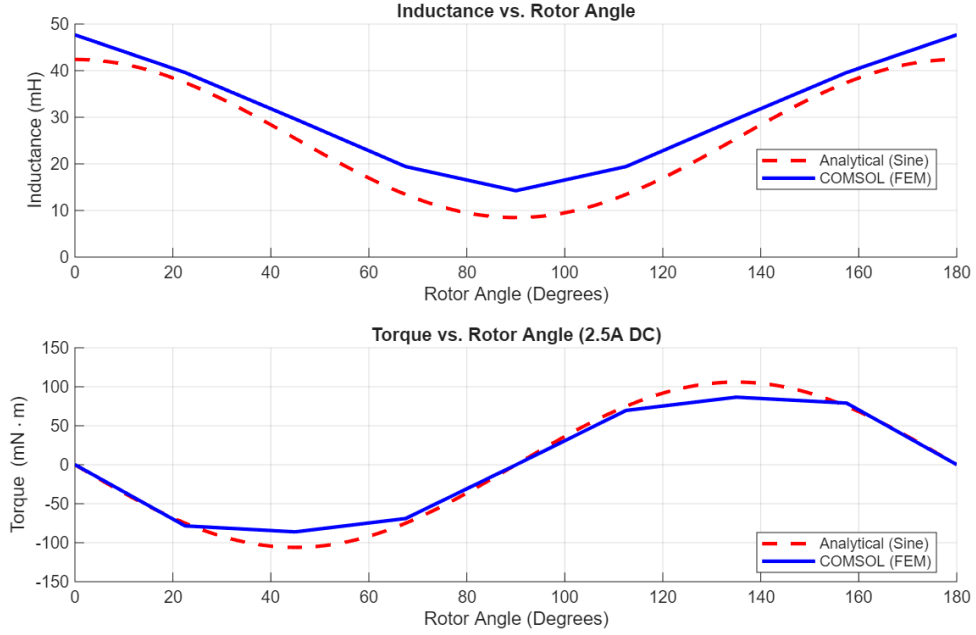


Figure 4: Comparison of analytical and FEM results (time-dependent) for the linear model.

## 4 FEA Modelling: 2D Non-linear Materials (Q3)

### 4.1 Flux Density Vectors (Q3-a)

To observe the effects of material saturation, the core material properties were updated to incorporate the actual B-H curve of Soft Iron, rendering the material non-linear. The

flux density vectors were replotted for 0, 45, and 90 degrees, as shown in Figure 5, Figure 6, and Figure 7, respectively.

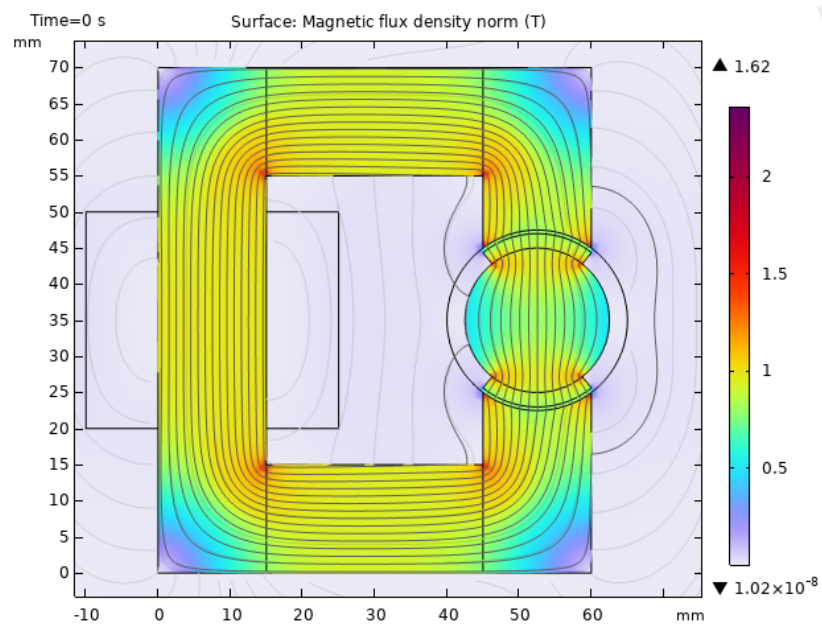


Figure 5: Flux density vectors at 0 degrees for non-linear soft iron.

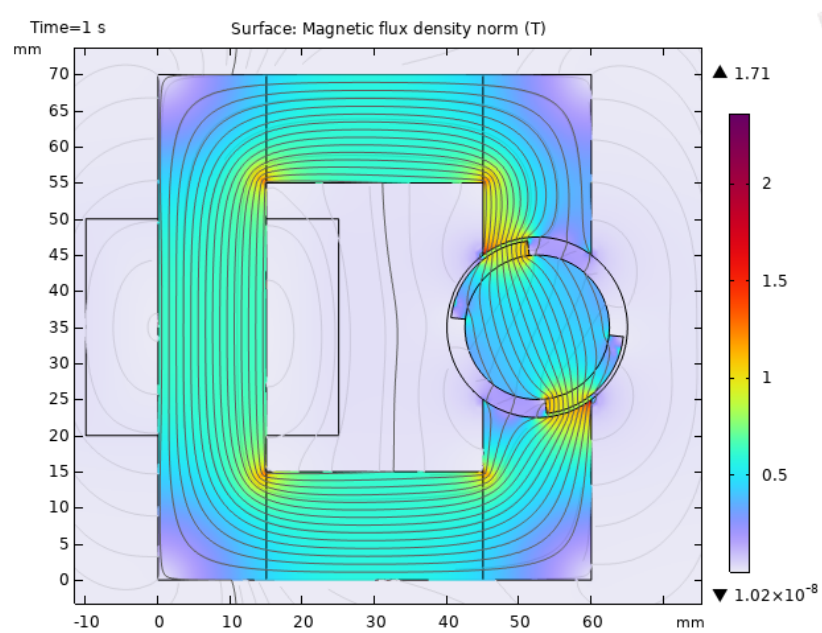


Figure 6: Flux density vectors at 45 degrees for non-linear soft iron.

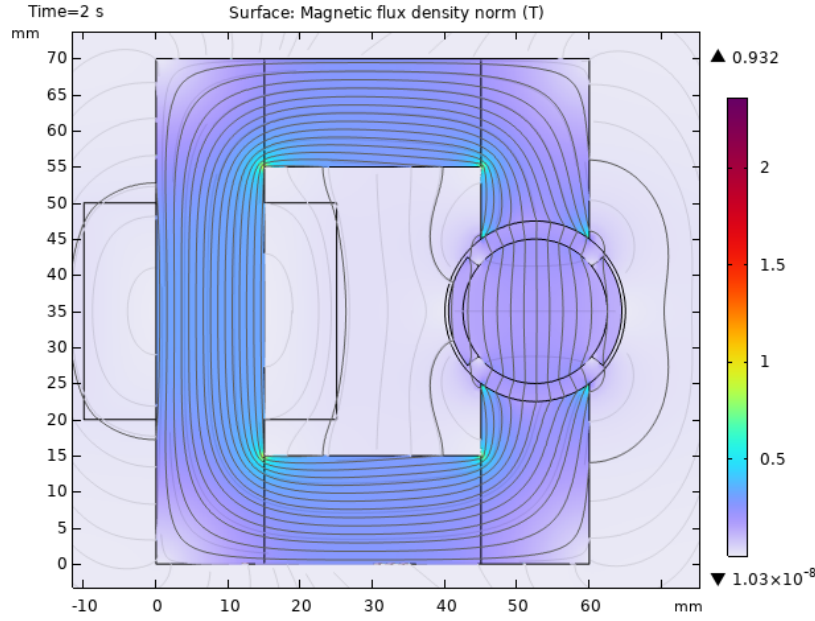


Figure 7: Flux density vectors at 90 degrees for non-linear soft iron.

## 4.2 Inductance and Stored Energy (Q3-b)

Unlike the linear analysis, a time-dependent dynamic study was performed on the non-linear model to accurately capture the system's response. At the fully aligned position (0 degrees), the stored energy was calculated as 0.135 J with an inductance of 0.0433 H. Compared to the linear model (0.0477 H), this inductance value is measurably lower due to magnetic saturation heavily reducing the permeability of the core material near the pole tips.

## 4.3 Torque Comparison with Analytical Model (Q3-c)

The continuous torque profile obtained from the non-linear time-dependent simulation was documented to compare directly with the linear conditions. The generated torque values at ten discrete positions are summarized in Table 2.

Table 2: Axial torque values at discrete rotor positions for the non-linear material model.

| Rotor Angle (°) | Axial Torque (mN·m) |
|-----------------|---------------------|
| 0.0             | 0.12                |
| 22.5            | -68.57              |
| 45.0            | -77.70              |
| 67.5            | -65.18              |
| 90.0            | -0.23               |
| 112.5           | 65.78               |
| 135.0           | 78.18               |
| 157.5           | 69.04               |
| 180.0           | 0.11                |
| 360.0           | 0.12                |

These FEM results were plotted against the initial analytical derivation, as illustrated in Figure 8.

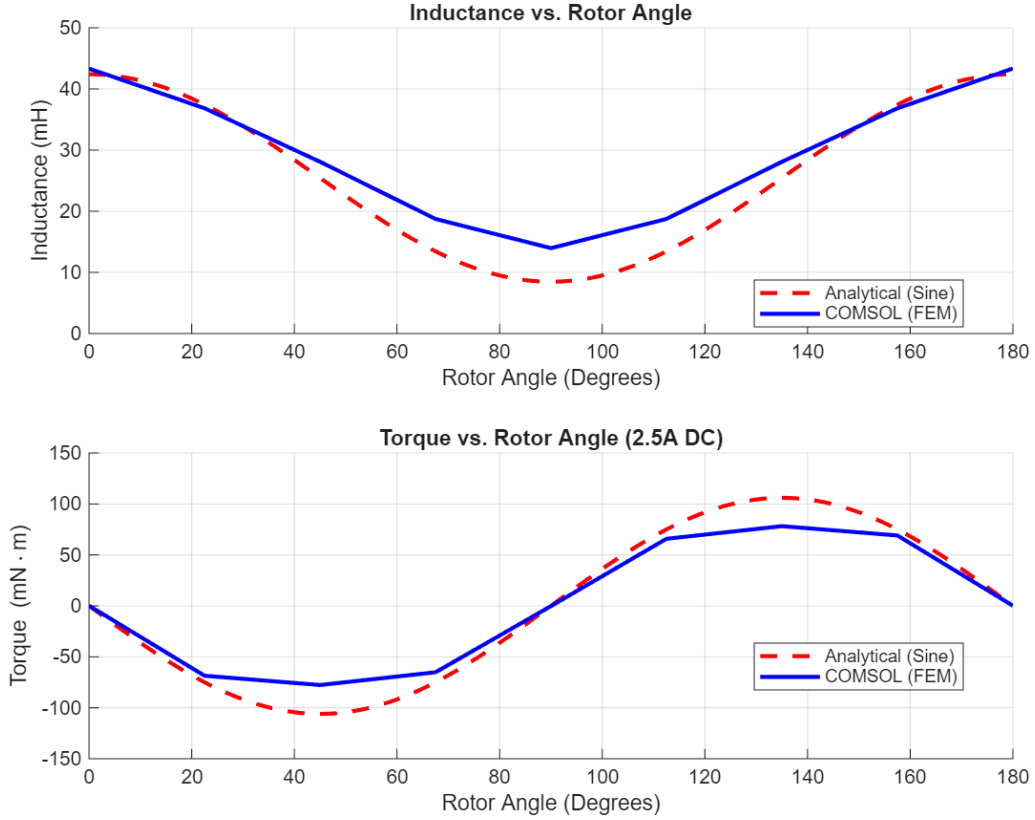


Figure 8: Comparison of analytical and non-linear FEM torque results.

The graph clearly displays the profound dampening effect of the physical realities on the machine's performance. While the purely analytical equation forecasts a peak torque of 106 mN·m, the introduction of non-linear material properties in the dynamic FEM model restricts the actual peak torque to 78.18 mN·m. The shape of the FEM torque curve also noticeably deviates from the perfect analytical sine wave, exhibiting wider bases and flattened peaks due to position-dependent saturation characteristics.

#### 4.4 Effects of Fringing and Saturation (Q3-d)

Comparing the analytical, linear, and non-linear results highlights the compounding impacts of fringing fields and magnetic saturation on torque production. The reluctance torque is fundamentally driven by the rate of change of inductance concerning position. Fringing fields, which occur when flux leaves the high-permeability iron path to bulge through the surrounding air, are most dominant at unaligned positions (e.g., 90 degrees). This phenomenon effectively increases the cross-sectional area of the air gap, raising the minimum inductance limit compared to idealized models.

Conversely, saturation occurs predominantly at aligned positions (0 and 180 degrees) where the magnetic flux is highly concentrated. The non-linear soft iron struggles to accommodate the increasing flux density, leading to a sharp drop in relative permeability. To illustrate this, local flux density measurements reveal that while the idealized linear model permits extreme concentrations up to 2.36 T at the sharp inner corners and 1.06



T at the core center, the physical non-linear model caps these values at 1.4 T and 0.97 T, respectively. This saturation effect heavily restricts the maximum achievable inductance. Because torque production relies on maximizing the difference between aligned and un-aligned inductances, the simultaneous increase of minimum inductance due to fringing and the suppression of maximum inductance due to saturation severely compresses the actual torque output of the physical machine.

## 5 Control Method (Q4)

To achieve continuous rotation, a position-dependent On/Off DC current excitation method is proposed. The torque production principle of a variable reluctance machine dictates that positive torque is generated only when the rotor is moving from an un-aligned position towards an aligned position, corresponding to regions where inductance is increasing. The extracted data confirms negative torque between 0 and 90 degrees, and positive torque between 90 and 180 degrees. Therefore, a rotor position sensor can be utilized to supply the 2.5 A DC current solely when the rotor is positioned between 90-180 degrees and 270-360 degrees. By completely cutting off the excitation current during the intervals where the inductance is decreasing, the generation of negative torque is effectively prevented, ensuring a strictly positive average torque and stable continuous rotation.

## 6 Motion Animation (Bonus-I)

To visualize the dynamic operation of the variable reluctance machine, rotation animations were generated. The GIF files showcasing the time-dependent rotation and flux density variations for both linear and non-linear cases are available in the project's digital repository. The animations can be viewed interactively via the linked GitHub repository detailed in Appendix B. These visuals provide an intuitive understanding of how the magnetic field distribution dynamically aligns the rotor with the stator poles.

## 7 Conclusion

This project successfully modeled and analyzed the torque characteristics of a variable reluctance machine. The transition from a simplified analytical approach to a linear simulation, and finally to a non-linear time-dependent dynamic analysis, demonstrated the critical importance of physical factors. The comparison graphs and evaluation data conclusively proved that analytical models overestimate performance, while non-linear FEA provides a realistic operational baseline by correctly interpreting the restrictive impacts of magnetic saturation and fringing flux. Furthermore, the proposed position-dependent On/Off control method verified the theoretical requirements for practical, continuous positive torque production.

## A Material Properties

The magnetic core was modeled using "Soft Iron" for both the stator and the rotor. For the linear analysis, a constant relative permeability ( $\mu_r$ ) of 2500 was assumed. For the non-linear FEA analysis, the intrinsic B-H magnetization curve for Soft Iron was utilized within the COMSOL material library. Figure 9 illustrates this non-linear relationship.

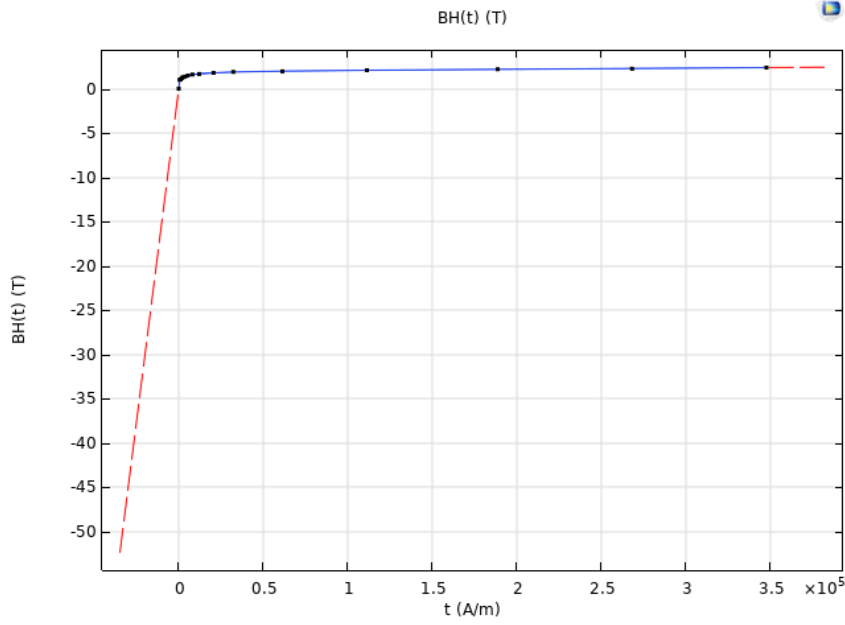


Figure 9: B-H magnetization curve for the Soft Iron core material.

This curve accounts for the core saturation effect at high magnetic flux densities. To define this material behavior in the simulation, discrete data points mapping the magnetic field intensity ( $H$ ) to the magnetic flux density ( $B$ ) were incorporated, specific values of which can be observed at critical saturation thresholds (e.g., reaching 1.5 T at 5425 A/m and 2.0 T at 61213 A/m). The exact data points utilized for the B-H curve are detailed in Table 3.

Table 3: B-H Curve Data Points for Soft Iron.

| Magnetic Field Intensity, $H$ (A/m) | Magnetic Flux Density, $B$ (T) |
|-------------------------------------|--------------------------------|
| 0.0                                 | 0.0                            |
| 663.15                              | 1.0                            |
| 1067.50                             | 1.1                            |
| 1705.23                             | 1.2                            |
| 2463.11                             | 1.3                            |
| 3841.67                             | 1.4                            |
| 5425.74                             | 1.5                            |
| 7957.75                             | 1.6                            |
| 12298.30                            | 1.7                            |
| 20462.80                            | 1.8                            |
| 32169.60                            | 1.9                            |
| 61213.40                            | 2.0                            |
| 111408.00                           | 2.1                            |
| 188487.76                           | 2.2                            |
| 267930.36                           | 2.3                            |
| 347507.84                           | 2.4                            |

## B GitHub Repository

All simulation files, analytical MATLAB scripts, extracted numerical datasets, and rotation animations are privately available on GitHub. The repository can be accessed via the following link:

github: metinozdemir01/EE568